Circles

1. Complete the square: write each expression as $(x+p)^2+q$.

1(a) $x^2+10x+28$

1(a)

1(b) x^2-6x+1

1(b)

1(c) x^2-12x

1(c)

1(d) x^2+7x

1(d)

2. Find the line through each pair of points.

2(a) A$(0,-6)$, B$(4,3)$

2(a)

2(b) P$(7,-5)$, Q$(-9,3)$

2(b)

2(c) R$(-4,-2)$, T$(5,10)$

2(c)

3. Use the discriminant to determine the number of real solutions.

3(a) $x^2-7x+14=0$

3(a)

3(b) $x^2+11x+8=0$

3(b)

3(c) $4x^2+12x+9=0$

3(c)

4 Find the line through $(3,-4)$ perpendicular to $6x-5y-1=0$.

4

1. Circle equations

Find centres, radii and equations.

2. Lines and circles

Find intersections and identify tangency.

3. Chords and tangents

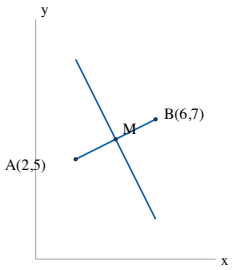
Use perpendicular bisectors and circle geometry.

4. Triangles in circles

Find the circle through three vertices.

Midpoints and Perpendicular Bisectors 4

A perpendicular bisector passes through the midpoint of a segment and meets it at a right angle.



1 Find the midpoint of A(2,5) and B(6,7).

1

2 Find the gradient of the perpendicular bisector.

2

3 Find its equation.

3

The perpendicular bisector of a chord passes through the centre of its circle.

Test Your Understanding 5

a Find the perpendicular bisector of A(4,7), B(10,17).

a

b Find a general perpendicular-bisector equation for distinct points $A(x_1, y_1), B(x_2, y_2)$.

b

c AB is a diameter with centre (5,−4). A=(1,−2). Find B.

c

Derive the bisector from its midpoint and perpendicular direction; there is no need to memorise the general expression.

Exercise 6A - Y1 Page 115 6

6 JK is a diameter with $J(\frac{3}{4}, \frac{4}{3})$, $K(-\frac{1}{2}, 2)$. The centre lies on $y = 8x + b$. Find b.

6

7 AB is a diameter with A(0,−2), B(6,−5). Show its centre lies on x−2y−10=0.

7

2 F$(-2,5)$, G$(2,9)$ are diameter endpoints. Find the line through the centre perpendicular to FG.

2

3 J$(0,-3)$, K$(4,-5)$ are diameter endpoints. Find the perpendicular line through the centre in integer standard form.

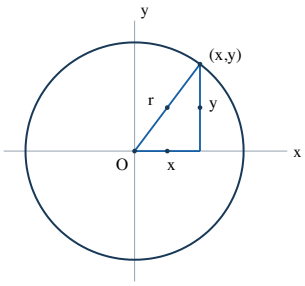
3

5 X$=(7,-2)$, Y$=(4,q)$. Their perpendicular bisector is $y=4x+b$. Find q and b.

5

Equation of a circle 9

A point (x, y) is on a circle centred at the origin with radius r. Draw a right-angled triangle from the origin to the point.

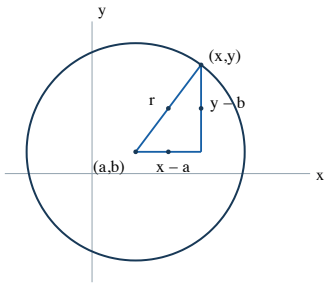


Equation What equation must x and y satisfy?

Equation

Equation of a circle 10

The centre moves to (a, b). The horizontal and vertical displacements to (x, y) are $x - a$ and $y - b$.



Equation Use Pythagoras to find the equation.

Equation

Quickfire Questions 11

Find the missing equation, or the centre and radius, for each circle.

a Centre
$(0, 0)$
Radius
5

a

b Centre
$(1, 2)$
Radius
6

b

c
$(x + 3)^2 + (y - 5)^2 = 1$

c

d
$(x + 5)^2 + (y - 2)^2 = 49$

d

e
$(x + 6)^2 + y^2 = 16$

e

f
$(x - 1)^2 + (y + 1)^2 = 3$

f

g
$(x + 2)^2 + (y - 3)^2 = 8$

g

Finding the equation using points 12

AB is a diameter with A(5,8), B(−7,4). Find the circle equation.

1 Find the centre.

1

2 Find the squared radius.

2

3 Write the equation.

3

Test Your Understanding 13

A=(5,−1), B=(13,11). AB is a diameter of circle C.

a Find the midpoint of AB.

a

b Find an equation for C.

b

Completing the square 14

Find the centre and radius of
$x^2 + y^2 - 6x + 2y - 6 = 0$

1 Group the x and y terms, then complete both squares.

1

2 Read the centre and radius from the standard form.

2

Further Example 15

A circle has equation
$x^2 + y^2 - 20x - 16y + 139 = 0$

a Find its centre.

a

b Show its radius is 5.

b

Example 16

Derive the centre and radius of
$x^2 + y^2 + ax + by + c = 0$

1 Complete the square.

1

2 Identify the centre and radius.

2

A real circle requires the expression under the square root to be positive. Zero gives a single point; a negative value gives no real points.

5 PQ is a diameter, with P(5,6) and Q(−2,2). Find the circle equation.

5

6 The point (1,−3) lies on $(x-3)^2+(y+4)^2=r^2$. Find r.

6

Complete the square, then find each centre and radius.

8(a)

$x^2+y^2-4x-11=0$

8(a)

9(a)

$x^2+y^2-10x+4y-20=0$

9(a)

8(b) Read the centre and radius.

8(b)

9(b) Read the centre and radius.

9(b)

12 P(7,−14) is on $x^2 + y^2 + 6x - 14y = 483$. PQ is a diameter. Find Q.

12

13 The circle $(x - k)^2 + y^2 = 41$ passes through (3,4). Find both values of k.

13

Extension

1 [MAT 2009 1B] Find the shortest distance from the origin to $x^2 + y^2 + 6x + 8y = 75$.

1

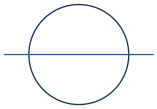
2 [MAT 2016 1I] For positive a,b and $x^2 + y^2 \leq 1$, find the greatest possible value of ax+by.

2

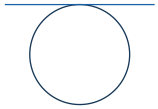
3 [MAT 2007 1D] Which point on $(x - 5)^2 + (y - 4)^2 = 4$ is closest to $(x - 1)^2 + (y - 1)^2 = 1$?

3

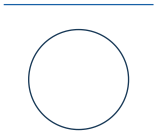
Solve a line and circle simultaneously by substitution. The resulting quadratic has two, one or no real roots: a secant, a tangent or no intersection.



Secant: two intersections



Tangent: one intersection



No intersection

Example Show that $y = x + 3$ never intersects $x^2 + y^2 = 1$.

Example

1 Find where $y = x + 6$ meets $x^2 + (y - 3)^2 = 29$.

1

2 Use algebra to find k when $y = x + k$ touches $x^2 + y^2 = 1$.

2

6. The line $x + y = 11$ meets the circle $x^2 + (y - 3)^2 = 32$.

a Show that there is only one intersection.

a

b Find the point of intersection.

b

7. The line $y = 2x - 2$ meets $(x - 2)^2 + (y - 2)^2 = 20$ at A and B.

7(a) Find A and B.

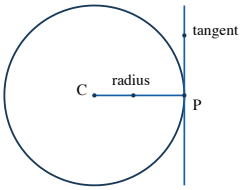
7(a)

7(b) Show that AB is a diameter.

7(b)

11 Find the possible k for which $y = 4x - 1$ does not meet the circle $x^2 + 2x + y^2 = k$.

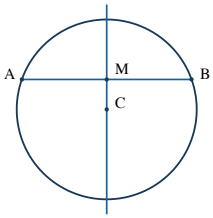
11



Tangent and radius

The tangent at a point on a circle is perpendicular to the radius to that point.

Find the radius gradient, take its negative reciprocal, then use the point to find the tangent equation.



Chord and centre

The perpendicular bisector of any chord passes through the centre.

Find the perpendicular bisectors of two chords. Their intersection is the circle centre.

Examples 26

1. Circle C: $(x-3)^2+(y-7)^2=100$, with P(11,1).

1(a) Verify P is on C.

1(a)

1(b) Find the tangent at P.

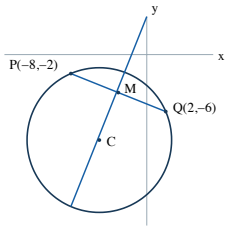
1(b)

2. Find both tangents of gradient -3 to $(x-4)^2+(y+4)^2=10$.

2

Determining the Circle Centre 27

P(−8,−2) and Q(2,−6) lie on a circle with centre C. M is the midpoint of PQ. The line l passes through M and C.



a Find an equation of l.

a

b(i) Given that the y-coordinate of C is −9, show its x-coordinate is −5.

b(i)

b(ii) Find the circle equation.

b(ii)

Test Your Understanding 28

1 Find the tangent at P(6,9) to the circle with centre C(3,5). Use integer standard form.

1

2 A circle passes through A(0,0), B(4,2). Its centre has x-coordinate −1. Find its equation.

2

6. The tangent to $(x+4)^2+(y-1)^2=242$ at $(7,-10)$ meets the y-axis at S and the x-axis at T.

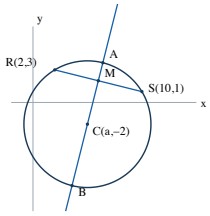
a Find S and T.

a

b Find the area of triangle OST, where O is the origin.

b

10. R(2,3), S(10,1) lie on a circle with centre C(a,−2). M is the midpoint of RS. The line l passes through M and C.



a Find l.

a

b Find a.

b

c Find the circle equation.

c

d Find where l meets the circle.

d

1 [MAT 2012 1A] Which line is tangent to $x^2 + y^2 = 4$?

A: $x + y = 2$

B: $y = x - 2\sqrt{2}$

C: $x = \sqrt{2}$

D: $y = \sqrt{2} - x$

1

2. [**AEA 2006 Q4**]

$y = mx$ is tangent to $(x + 4)^2 + (y - 7)^2 = 13$.

(a) Show $3m^2 + 56m + 36 = 0$.

(b) Find the two contact points A and B.

(c) Tangents from $(4,-7)$ touch $x^2 + y^2 = 13$ at P and Q. Find P or Q.

Worked solution: next slide.

3. [**STEP 2005 Q6**]

A=(5,16), B=(−4,4), and AP=2BP.

(i) Show the locus of P is $(x + 7)^2 + y^2 = 100$.

(ii) C=(a,0), D=(b,0), QC=kQD, k>1. This gives the same locus. Show

$$\frac{a+7}{b+7} = \frac{a^2+51}{b^2+51}.$$

For a≠b, show $(a + 7)(b + 7) = 100$.

Worked solution: slide 33.

[AEA 2006 Q4] Tangents from the origin to $(x + 4)^2 + (y - 7)^2 = 13$.

a Find the gradient condition.

a

b Find the contact points.

b

c Find a contact point for tangents from $(4,-7)$ to $x^2 + y^2 = 13$.

c

[STEP 2005 Q6] A=(5,16), B=(−4,4), AP=2BP.

i Find the locus.

i

ii Compare the locus QC=kQD with the same circle.

ii

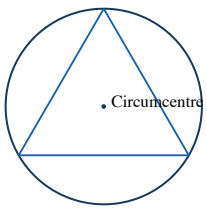
Conclusion Deduce the result when $a \neq b$.

Conclusion

Triangles in Circles 34

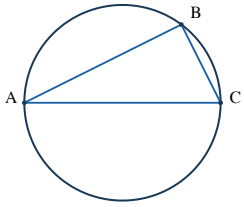
A triangle is inscribed in a circle when all three vertices lie on the circle.

The circle circumscribes the triangle. It is called the circumcircle; its centre is the circumcentre.



If $\angle ABC = 90°$, AC is the diameter of the circumcircle.

Conversely, if AC is a diameter and B is on the circle, $\angle ABC = 90°$. Thus AB is perpendicular to BC and $AB^2 + BC^2 = AC^2$.



To find the circumcentre of any triangle: find the perpendicular bisectors of two sides, then solve for their intersection.

Example 36

A=(−8,1), B=(4,5), C=(−4,9).

a: method 1 Show AB is a diameter using Pythagoras.

a: method 1

a: method 2 Show the two sides are perpendicular.

a: method 2

b Find the circle equation.

b

Example

Find the circle through A(0,2), B(2,0) and C(8,18).

1 Find two perpendicular bisectors.

1

2 Find the centre and radius.

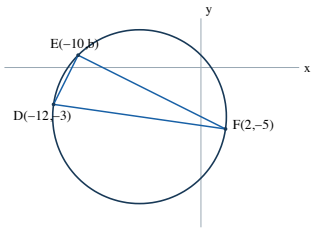
2

3 Write the circle equation.

3

Exercise 6F - Y1 Page 131-132

8. D(−12,−3), E(−10,b), F(2,−5) lie on a circle. Given $\angle DEF = 90°$ and b>0:



a Show b=1.

a

b Find the circle equation.

b

9. A circle has equation $x^2 + 2x + y^2 - 24y - 24 = 0$.

a Find its centre and radius.

a

b A$(-13,17)$ and B$(11,7)$ lie on the circle. Show AB is a diameter.

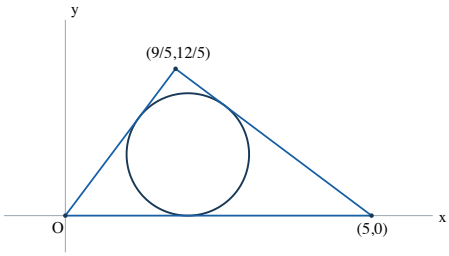
b

c C lies on the negative x-axis and angle ACB is a right angle. Find C.

c

[STEP] A circle touches all three sides of the triangle bounded by $y = 0$, $3y = 4x$ and $4y + 3x = 15$. Find its equation.

Given: the family $(x - 2t)^2 + (y - t)^2 = t^2$, t>0, touches the first two lines. The incircle lies entirely inside the triangle.



1 Use the first two sides.

1

2 Use the third side.

2

3 Write the equation.

3